

Practical 7

Write a program that prints the addresses of code, global, static, heap, and stack variables. Analyze the Linux process address space layout and explain each memory segment.

Make use of the `/proc//maps` file and memory analysis tools to study the memory organization of a running process. Prepare a report explaining virtual memory mappings.

```
#include <stdio.h>
```

```
#include <stdlib.h>
```

```
#include <unistd.h>
```

```
int global_var = 100;
```

```
int global_uninit;
```

```
static int static_var = 200;
```

```
static int static_uninit;
```

```
void sample_function()
```

```
{
```

```
    printf("Code address: %p\n", (void *)sample_function);
```

```
}
```

```
int main()
```

```
{
```

```
    int stack_var = 10;
```

```
    static int static_local = 300;
```

```
    int *heap_var = malloc(sizeof(int));
```

```
    *heap_var = 400;
```

```
printf("PID: %d\n", getpid());
```

```
sample_function();
```

```
printf("Global variable: %p\n", (void *)&global_var);
```

```
printf("Uninitialized global: %p\n", (void *)&global_uninit);
```

```
printf("Static variable: %p\n", (void *)&static_var);
```

```
printf("Uninitialized static: %p\n", (void *)&static_uninit);
```

```
printf("Static local: %p\n", (void *)&static_local);
```

```
printf("Heap variable: %p\n", (void *)heap_var);
```

```
printf("Stack variable: %p\n", (void *)&stack_var);
```

```
printf("\nProcess running. Press Enter to exit...\n");
```

```
getchar();
```

```
free(heap_var);
```

```
return 0;
```

```
}
```

```

ganesh@Ganesh:~$ nano practical7.c
ganesh@Ganesh:~$ gcc practical7.c -o practical7
ganesh@Ganesh:~$ ./practical7
PID: 545
Code address: 0x5698354b2209
Global variable: 0x5698354b5010
Uninitialized global: 0x5698354b5020
Static variable: 0x5698354b5014
Uninitialized static: 0x5698354b5024
Static local: 0x5698354b5018
Heap variable: 0x56985e171010
Stack variable: 0x7ffc8ec0153c

Process running. Press Enter to exit...

```

```

ganesh@Ganesh:~$ cat /proc/545/maps
5698354b1000-5698354b2000 r--p 00000000 08:30 14505 /
me/snigdha/practical7
5698354b2000-5698354b3000 r-xp 00001000 08:30 14505 /
me/snigdha/practical7
5698354b3000-5698354b4000 r--p 00002000 08:30 14505 /
me/snigdha/practical7
5698354b4000-5698354b5000 r--p 00002000 08:30 14505 /
me/snigdha/practical7
5698354b5000-5698354b6000 rw-p 00003000 08:30 14505 /
me/snigdha/practical7
56985e171000-56985e192000 rw-p 00000000 00:00 0 [
ap]
7d887b400000-7d887b428000 r--p 00000000 08:30 13641 /
r/lib/x86_64-linux-gnu/libc.so.6
7d887b428000-7d887b5c0000 r-xp 00028000 08:30 13641 /
r/lib/x86_64-linux-gnu/libc.so.6
7d887b5c0000-7d887b60e000 r--p 001c0000 08:30 13641 /
r/lib/x86_64-linux-gnu/libc.so.6
7d887b60e000-7d887b612000 r--p 0020d000 08:30 13641 /
r/lib/x86_64-linux-gnu/libc.so.6
7d887b612000-7d887b614000 rw-p 00211000 08:30 13641 /
r/lib/x86_64-linux-gnu/libc.so.6
7d887b614000-7d887b621000 rw-p 00000000 00:00 0
7d887b808000-7d887b80b000 rw-p 00000000 00:00 0
7d887b810000-7d887b812000 rw-p 00000000 00:00 0
7d887b812000-7d887b816000 r--p 00000000 00:00 0 [
ar]
7d887b816000-7d887b818000 r--p 00000000 00:00 0 [
ar_vclock]
7d887b818000-7d887b81a000 r-xp 00000000 00:00 0 [

```

```

ganesh@Ganesh:~$ pmap -x $PID
pmap: argument missing
ganesh@Ganesh:~$ pmap -x 545
545: ./practical7
Address          Kbytes      RSS    Dirty Mode  Mapping
00005698354b1000      4         4        0 r---- practical7
00005698354b2000      4         4        0 r-x-- practical7
00005698354b3000      4         4        0 r---- practical7
00005698354b4000      4         4        4 r---- practical7
00005698354b5000      4         4        4 rw--- practical7
000056985e171000    132         4        4 rw--- [ anon ]
00007d887b400000    160        156        0 r---- libc.so.6
00007d887b428000   1632       1120        0 r-x-- libc.so.6
00007d887b5c0000    312        184        0 r---- libc.so.6
00007d887b60e000     16         16       16 r---- libc.so.6
00007d887b612000      8          8        8 rw--- libc.so.6
00007d887b614000     52         20       20 rw--- [ anon ]
00007d887b808000     12          8        8 rw--- [ anon ]
00007d887b810000      8          4        4 rw--- [ anon ]
00007d887b812000     16          0        0 r---- [ anon ]
00007d887b816000      8          0        0 r---- [ anon ]
00007d887b818000      8          8        0 r-x-- [ anon ]
00007d887b81a000      4          4        0 r---- ld-linux-x86-64.so.2
00007d887b81b000    188        184        0 r-x-- ld-linux-x86-64.so.2
00007d887b84a000     44         44        0 r---- ld-linux-x86-64.so.2
00007d887b855000      8          8        8 r---- ld-linux-x86-64.so.2
00007d887b857000      4          4        4 rw--- ld-linux-x86-64.so.2
00007d887b858000      4          4        4 rw--- [ anon ]
00007ffc8ebe2000    132         12       12 rw--- [ stack ]
-----
total kB          2768    1808     96
ganesh@Ganesh:~$ grep -E 'heap|stack|practical|libc|ld-linux' /proc/545/maps
5698354b1000-5698354b2000 r--p 00000000 08:30 14505 /home/snigdha/practical7
5698354b2000-5698354b3000 r-xp 00001000 08:30 14505 /home/snigdha/practical7

```

```
#include <stdio.h>
```

```
#include <stdlib.h>
```

```
#include <unistd.h>
```

```
int global_var = 10;
```

```
static int static_var = 20;
```

```
int main()
```

```
{
```

```
int stack_var = 30;

int *heap_var = malloc(sizeof(int));

*heap_var = 40;


printf("PID: %d\n", getpid());

printf("Code: %p\n", (void *)main);

printf("Global: %p\n", (void *)&global_var);

printf("Static: %p\n", (void *)&static_var);

printf("Heap: %p\n", (void *)heap_var);

printf("Stack: %p\n", (void *)&stack_var);


fflush(stdout);

sleep(60);


free(heap_var);

return 0;

}
```

```
ganesh@Ganesh:~$ nano practical7_live.c
ganesh@Ganesh:~$ gcc practical7_live.c -o practical7_live
ganesh@Ganesh:~$ ./practical7_live
PID: 823
Code: 0x55b6e4bab209
Global: 0x55b6e4bae010
Static: 0x55b6e4bae014
Heap: 0x55b70ca30010
Stack: 0x7ffdd937c9ec
ganesh@Ganesh:~$ |
```

```

ganesh@Ganesh:~$ cat /proc/823/maps
55b6e4baa000-55b6e4bab000 r--p 00000000 08:30 32039
me/snigdha/practical7_live
55b6e4bab000-55b6e4bac000 r-xp 00001000 08:30 32039
me/snigdha/practical7_live
55b6e4bac000-55b6e4bad000 r--p 00002000 08:30 32039
me/snigdha/practical7_live
55b6e4bad000-55b6e4bae000 r--p 00002000 08:30 32039
me/snigdha/practical7_live
55b6e4bae000-55b6e4baf000 rw-p 00003000 08:30 32039
me/snigdha/practical7_live
55b70ca30000-55b70ca51000 rw-p 00000000 00:00 0
ap]
76ffdea00000-76ffdea28000 r--p 00000000 08:30 13641
r/lib/x86_64-linux-gnu/libc.so.6
76ffdea28000-76ffdebc0000 r-xp 00028000 08:30 13641
r/lib/x86_64-linux-gnu/libc.so.6
76ffdebc0000-76ffdec0e000 r--p 001c0000 08:30 13641
r/lib/x86_64-linux-gnu/libc.so.6
76ffdec0e000-76ffdec12000 r--p 0020d000 08:30 13641
r/lib/x86_64-linux-gnu/libc.so.6

```

```

ganesh@Ganesh:~$ pmap -x 823
823:    ./practical7_live
Address      Kbytes      RSS      Dirty Mode  Mapping
000055b6e4baa000      4         4         0 r---- practical7_live
000055b6e4bab000      4         4         0 r-x-- practical7_live
000055b6e4bac000      4         4         0 r---- practical7_live
000055b6e4bad000      4         4         4 r---- practical7_live
000055b6e4bae000      4         4         4 rw--- practical7_live
000055b70ca30000    132         4         4 rw--- [ anon ]
000076ffdea00000    160        160         0 r---- libc.so.6
000076ffdea28000   1632       1184         0 r-x-- libc.so.6
000076ffdebc0000    312        184         0 r---- libc.so.6
000076ffdec0e000     16         16        16 r---- libc.so.6
000076ffdec12000      8          8         8 rw--- libc.so.6
000076ffdec14000     52         20        20 rw--- [ anon ]
000076ffdec96000     12          8         8 rw--- [ anon ]
000076ffdec9e000      8          4         4 rw--- [ anon ]
000076ffdeca0000     16          0         0 r---- [ anon ]
000076ffdeca4000      8          0         0 r---- [ anon ]
000076ffdeca6000      8          8         0 r-x-- [ anon ]
000076ffdeca8000      4          4         0 r---- ld-linux-x86-64.so.2
000076ffdeca9000    188       188         0 r-x-- ld-linux-x86-64.so.2
000076ffdecdd8000    44         44         0 r---- ld-linux-x86-64.so.2
000076ffdece3000      8          8         8 r---- ld-linux-x86-64.so.2
000076ffdece5000      4          4         4 rw--- ld-linux-x86-64.so.2
000076ffdece6000      4          4         4 rw--- [ anon ]
000076ffdd935e000   132         12        12 rw--- [ stack ]
-----
total kB          2768      1880      96
ganesh@Ganesh:~$ ps -p 823 -o pid,comm,vsz,rss
  PID COMMAND          VSZ   RSS

```